

Backend-focused software engineering student with high interest in developing scalable, high-performance services and managing complex database systems. Possessing strong proficiency in full-stack development, with additional expertise in Web3 and Artificial Intelligence.

Skills and Languages

1. Backend: NodeJS, Express, FastAPI, Flask, MongoDB, Postgres
2. Frontend: ReactJS, React Native, Tailwind
3. Blockchain: Solidity, Truffle, Ganache
4. Artificial Intelligence: TensorFlow, Keras, Deep Learning Models
5. Others: Linux, Git, Github, Azure

Education

- **B.Tech. CSE**, Bennett University, Greater Noida (8.8 GPA , 2023–2027, ongoing)

Work Experience

- **Software Development Intern** : Empty Cup Ltd., Remote Jun 2025 – Aug 2025
 - Implemented rate limiting on computationally intensive API endpoints within a complex Svelte-Flask monorepo codebase.
 - Developed features enabling collaborative work on shared projects for multiple users.
 - Integrated an automation layer for quotation and billing calculation.
 - Addressed and resolved software defects.

Projects

- **Blockchain Healthcare Record System** · Solidity, Truffle, MERN
 - Developed a blockchain-based system for secure storage and verification of patient health records.
 - Integrated Aadhar authentication and OTP verification using Firebase for user onboarding.
 - Designed a decentralized IPFS-backed storage system to fragment patient data across hospital nodes.
- **Redis-Based Key-Value Store in-Memory Database** · C++
 - Engineered a Redis server parser in C++ supporting GET, SET with PX expiration.
 - Focused on memory optimization and correct invalidation of expired keys in real-time.
- **BitTorrent Client Implementation** · JavaScript, Node.js
 - Built a simplified BitTorrent client capable of parsing .torrent files and downloading pieces from peers.
 - Implemented piece selection, block requests, and peer wire protocol handling for efficient downloads.
 - Gained deep understanding of distributed peer-to-peer file-sharing systems and network protocols.
- **Fullstack Multiplayer TypeRacer Game** · MERN Stack, WebSockets, MongoDB
 - Designed a real-time multiplayer typing game with live race updates using WebSocket communication.
 - Built user authentication, dynamic lobby creation, and race progress tracking.
 - Optimized server state management for handling multiple concurrent races smoothly.

Interests

- Currently exploring the world of Computer Graphics and Blockchain.
- 3d modelling on Blender